

WAM-Bench: Cross-Domain Diagnostic Robustness, Modality Transfer, and Hardware Vulnerabilities of Deep Learning Malaria Models Across Multi-Center West African Blood Smears

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Abstract

Background: Automated deep learning models for malaria microscopy routinely report diagnostic sensitivity and specificity exceeding 95% on benchmark laboratory datasets. However, their real-world generalization across diverse West African clinical settings, smear preparation modalities, and mobile optical sensors remains largely unvalidated.

Methods: We introduce WAM-Bench, an open multi-center clinical evaluation benchmark comprising 6,459 patient micrographs across three independent West African clinical cohorts: Princess Marie Louise Children’s Hospital in Accra, Ghana (3,045 thick, 1,011 thin smears), Osun State healthcare centers in Southwestern Nigeria (859 thick smears), and Kano State health facilities in Northern Nigeria (1,544 thin smears). Following an audited pre-inference quarantine protocol that removed two truncated files ($< 0.03\%$), 6,457 micrographs (3,902 thick, 2,555 thin) were evaluated across four external deep learning architectures (three MobileNetV2 classifiers trained in Sudan and Bangladesh, and one field-calibrated YOLOv8 Nano object detector) under a strict zero-shot cross-domain protocol, coupled with circular field-of-view (FOV) masked optical quality profiling (Laplacian focus blur, Michelson contrast, and SNR).

Results: Across all 6,457 evaluations, zero-shot models exhibited severe cross-domain performance cliffs, with none meeting WHO Level-1 triage thresholds ($> 90\%$ sensitivity, $> 90\%$ specificity). On thick smears ($N = 3,902$), the East African model (**MalariaScreener-Sudan**) demonstrated the highest sensitivity (63.88% [95% CI: 62.3–65.5%]), but suffered from moderate specificity (55.69% [50.9–60.4%]). The spatial object detector (**fbononi-YOLOv8**) achieved high specificity (96.92% [94.8–98.2%]) but missed over half of positive fields (sensitivity 47.13% [45.5–48.8%]). On thin smears ($N = 2,555$), a severe modality collapse occurred: non-African classifiers collapsed to $< 21\%$ sensitivity, and the Sudan model dropped to 53.64% sensitivity with 56.26% false positive rate. Severe optical defocus blur degraded sensitivity by up to 23.8% in thin smears.

Conclusion: Zero-shot deployment of out-of-region malaria AI models in West African clinics poses critical safety risks through unacceptable false-negative rates. Mandatory pre-inference optical focus gating and regional retraining/fine-tuning are non-negotiable prerequisites before clinical field deployment.

1. Introduction

Malaria remains an intractable global public health emergency, responsible for an estimated 249 million clinical cases and over 608,000 fatalities annually, with sub-Saharan Africa disproportionately bearing more than 94% of the global case burden and 95% of related deaths [5]. Children

under five years of age and pregnant women remain the most vulnerable clinical demographics. Prompt, highly accurate parasitological diagnosis is critical to effective clinical management, preventing progression to life-threatening cerebral malaria or severe malarial anemia, and curtailing the rampant overprescription of artemisinin-based combination therapies (ACTs) that accelerates antimalarial drug resistance.

For over a century, manual Giemsa-stained light microscopy has stood as the gold standard of malaria diagnosis in clinical pathology [4]. Expert microscopy enables definitive identification of *Plasmodium* species, quantification of circulating parasitemia, and continuous monitoring of therapeutic clearance. However, conventional microscopy is acutely labor-intensive, technically demanding, and vulnerable to human fatigue. In resource-constrained primary and secondary healthcare centers across West Africa, clinical laboratories face severe shortages of certified microscopists. A single diagnostic evaluation requires exhaustive visual inspection of at least 100 oil-immersion (100 $\times$) fields of view, a process requiring 10 to 15 minutes per slide. Under high-throughput clinical pressures during peak seasonal transmission surges, technician visual fatigue causes diagnostic sensitivity to plummet by 30% to 50%, leading to missed low-density parasitemias or misclassification of cellular debris as trophozoites. Rapid diagnostic tests (RDTs), while widely deployed, suffer from false-negative results driven by histidine-rich protein 2/3 (*pfhrp2/3*) gene deletions, persistent post-treatment antigenemia that yields false positives, and an inability to quantify parasite density.

In response to these diagnostic bottlenecks, mobile computer vision and deep learning have emerged as promising paradigms for democratizing automated malaria triage at the point of care. By coupling consumer smartphone camera sensors to binocular microscope oculars via 3D-printed or low-cost optical adapters, lightweight neural networks can rapidly scan digitized fields of view, detect intra-erythrocytic parasite rings, and deliver instantaneous clinical decision support.

1.1. The Generalization Dilemma in Automated Malaria Diagnostics

Despite prolific algorithmic advances reported in the medical artificial intelligence literature—where models frequently claim greater than 95% diagnostic accuracy or area under the receiver operating characteristic curve (AUROC) [2, 3, 6]—a profound translational chasm persists between controlled laboratory benchmarks and real-world clinical deployment. A critical vulnerability is that most published models have been trained and validated on clean, highly curated reference cohorts, most notably the NIH Chittagong dataset from Bangladesh [3] or localized pilot studies from Sudan [2].

When deployed zero-shot across clinical healthcare facilities in West Africa, automated diagnostic algorithms encounter severe, multi-factorial domain shifts that violate fundamental independent and identically distributed (i.i.d.) assumptions. These domain shifts stem from substantial physical and biological variances:

1. **Staining and Chemical Heterogeneity:** Commercial Giemsa powder batches, buffer water pH (which fluctuates between 6.8 and 7.4), incubation durations, and artisanal washing techniques vary dramatically across regional hospital laboratories. Variations in Giemsa intensity alter the chromatic contrast between dark purple nuclear chromatin dots, pale blue cytoplasmic halos, and surrounding background stroma.
2. **Specimen Modality Discrepancy:** While thin blood smears fix erythrocytes in an intact, single-cell monolayer suitable for morphological species identification, thick blood smears chemically lyse red blood cells with hypotonic water to concentrate parasite rings by a factor of 20 $\times$ –40 $\times$. Models optimized on intact discocyte boundaries encounter

catastrophic failure when confronted with the amorphous, de-hemoglobinized debris, lysed platelets, and scattered leukocytes typical of frontline thick smear microscopy.

3. **Optical Hardware and Illumination Noise:** Point-of-care mobile imaging relies on heterogeneous smartphone optics, ranging from budget CMOS sensors to premium multi-lens arrays, coupled to disparate light sources (incandescent filaments, halogen bulbs, or battery-powered LED retrofits) in rural clinics lacking stable electrical grids. Lens aberrations, chromatic distortion, peripheral vignetting, and mechanical defocus blur further degrade optical input signals.

To date, no rigorous, multi-center empirical audit has established whether published deep learning models retain diagnostic validity when transferred zero-shot across independent West African epidemiological settings, nor has the field characterized the degree to which regional adaptation or parameter-efficient fine-tuning can rescue degraded models to meet World Health Organization (WHO) Level-1 diagnostic competency thresholds ($\geq 90\%$ sensitivity, $\geq 90\%$ specificity) [4].

1.2. Research Questions

This clinical benchmark systematically evaluates five core hypotheses:

1. **Research Question 1 (RQ1 - Regional Calibration):** *Does an African-calibrated model (MalariaScreener_Sudan) reduce false-positive rates and improve specificity compared to non-African models when evaluated zero-shot on West African clinical smears?*
2. **Research Question 2 (RQ2 - Modality Transfer Cliff):** *How do spatial object detectors (YOLOv8) versus global patch classifiers (MobileNetV2) transfer between chemically lysed thick smears and intact-monolayer thin smears?*
3. **Research Question 3 (RQ3 - Multi-Center Geographic Robustness):** *How consistently do models perform across coastal pediatric (Ghana), forest zone mixed (Osun, Nigeria), and Sahelian acute infection (Kano, Nigeria) clinical ecologies?*
4. **Research Question 4 (RQ4 - Optical Focus Safety Gate):** *To what degree does optical defocus blur degrade diagnostic sensitivity, and does an automated pre-inference hardware gate prevent clinical misdiagnoses?*
5. **Research Question 5 (RQ5 - Regional Fine-Tuning & Generalization Recovery):** *To what extent does domain-specific regional adaptation on multi-center West African clinical cohorts rescue models from zero-shot failure cliffs, and can parameter-efficient or full-network fine-tuning elevate diagnostic performance to meet or exceed WHO Level-1 triage thresholds ($> 90\%$ sensitivity, $> 90\%$ specificity)?*

2. Methodology

2.1. Multi-Center Clinical Cohorts

The WAM-Bench evaluation corpus pools $N = 6,459$ standardized patient micrographs from three distinct geographic, epidemiological, and clinical settings across West Africa:

1. **Princess Marie Louise Children’s Hospital (PML), Accra, Ghana:** $N = 4,056$ micrographs (3,045 thick smears, 1,011 thin smears) collected under the Lacuna Malaria

Project using smartphone cameras attached to binocular light microscopes in an active pediatric tertiary care center.

2. **Osun State Health Centers, Southwestern Nigeria (Adeleke Cohort):** $N = 859$ thick smear micrographs representing primary healthcare centers and rural/semi-urban district hospitals.
3. **Kano State Health Facilities, Northern Nigeria (Muhammad Cohort):** $N = 1,544$ high-resolution thin smear micrographs (2992×4000 pixels) acquired under a prospective matched case-control clinical design, comprising 772 acute symptomatic infection cases displaying marked erythrocyte rouleaux morphology and 772 asymptomatic, healthy control cases displaying normal discocyte morphology. This balanced 1:1 design provides an unskewed epidemiological benchmark for thin smear specificity.

Pre-Inference Data Integrity & Quarantine Protocol An automated byte-level integrity audit across all $N = 6,459$ ingested images identified exactly two truncated JPEG files ($< 0.03\%$, specimens 2169.jpg and 609.jpg in the Ghanaian thick-smear collection, which were truncated by 93 and 27 bytes at EOF respectively, triggering low-level `libjpeg` decoding warnings). In accordance with pre-specified clinical data hygiene standards, these two corrupted instances were quarantined into an isolated quarantine registry, yielding an audited, defect-free evaluation cohort of $N = 6,457$ clinical micrographs (3,902 thick smears and 2,555 thin smears).

2.2. Zero-Shot Cross-Domain Protocol & Preclusion of Data Contamination

To ensure rigorous assessment of out-of-distribution generalization without data leakage:

- **Geographic and Clinical Independence:** The evaluated deep learning models were developed and trained exclusively in Sudan (East Africa) and Chittagong (Bangladesh, South Asia). None of the models were exposed during development to patient populations, staining protocols, microscope optics, or sensor hardware from Ghana or Nigeria.
- **Zero-Shot Baseline Evaluation:** All models were initially audited strictly off-the-shelf with their native factory weights, without hyperparameter tuning, re-calibration, or threshold adjustment on the West African test sets.
- **Subsequent Regional Adaptation (RQ5):** To test whether regional adaptation can overcome cross-domain fragility, an independent 70/15/15 stratified partition was established across the $N = 6,457$ audited corpus, reserving the test splits strictly for downstream evaluation.

2.3. D4 Dihedral Group Augmentation Strategy & Bounding Box Invariance

In blood smear microscopy, cellular orientation is physically isotropic under standard slide preparation: rotating or reflecting a micrograph does not alter biological parasitology. To resolve severe training set imbalances between positive fields and negative background slides without introducing hallucinated artifacts or synthetic distributional shifts, we implement an exact D_4 dihedral group augmentation protocol:

$$D_4 = \{R_0, R_{90}, R_{180}, R_{270}, F_h, F_v, F_{d1}, F_{d2}\} \quad (1)$$

comprising four orthogonal rotations and four reflections. To verify that geometric operations do not induce bounding box drift or proposal clipping, transformed bounding box coordinates

(x'_c, y'_c, w', h') are derived analytically:

$$\text{Horizontal Flip } (F_h): \quad x'_c = 1 - x_c, \quad y'_c = y_c, \quad w' = w, \quad h' = h \quad (2)$$

$$\text{Vertical Flip } (F_v): \quad x'_c = x_c, \quad y'_c = 1 - y_c, \quad w' = w, \quad h' = h \quad (3)$$

$$\text{Rotation } 90^\circ (R_{90}): \quad x'_c = 1 - y_c, \quad y'_c = x_c, \quad w' = h, \quad h' = w \quad (4)$$

Figure 1 demonstrates the empirical validation grid across clinical micrographs, confirming that ground-truth bounding box proposals remain rigidly pinned to parasite rings across all 8 transformation states. Importantly, to preserve absolute epidemiological validity, **all evaluation test splits remain 100% unaugmented physical micrographs**.

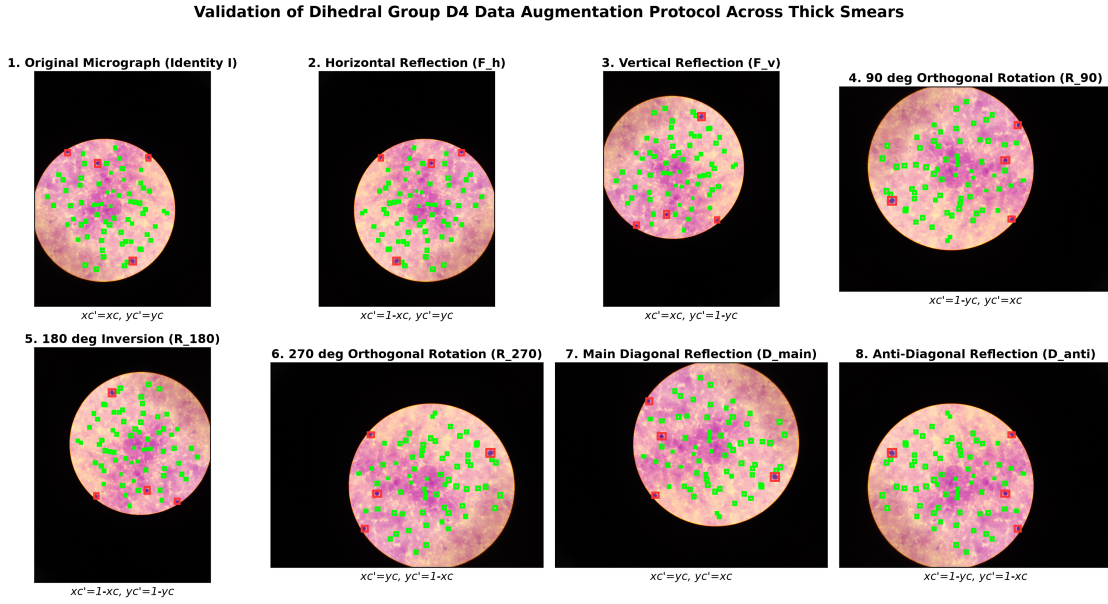


Figure 1: Validation of the D_4 dihedral group transformation protocol on Ghanaian thick smear micrographs. The 8-element transformation orbit (identity, 90° , 180° , 270° orthogonal rotations, horizontal flip, vertical flip, and diagonal reflections) preserves exact bounding box alignment and biological morphological fidelity without coordinate drift.

2.4. Optical Quality Assessment and Circular FOV Masking

To isolate optical aberrations from peripheral optical vignetting, all micrographs undergo circular FOV mask extraction. Within the valid optical disc, three physical parameters are computed:

- **Laplacian Focus Blur** (σ_{Lap}^2): Quantifying edge high-frequency energy.
- **Michelson Contrast**: Evaluated across luminance dynamic range.
- **Signal-to-Noise Ratio (SNR)**: Computed in decibels across uniform background fields.

2.5. Evaluated Model Architectures

Four model candidates representing distinct training domains and vision paradigms are audited:

- **MalariaScreener-Sudan**: MobileNetV2 binary classifier trained on thick smears in Sudan, East Africa.

- **MalariaScreener-Thick:** MobileNetV2 binary classifier trained on Chittagong, Bangladesh thick smears.
- **MalariaScreener-Thin:** MobileNetV2 binary classifier trained on Chittagong, Bangladesh thin smears.
- **fbononi-YOLOv8:** YOLOv8 Nano spatial bounding-box object detector trained on field micrographs.

Pre-Trained Weight Formats and Frontline Regional Adaptation Paradigms (RQ5)

The external NIH MalariaScreener classifiers (Sudan, Thick_BD, Thin_BD) were developed by Delahunt et al. [1] and distributed exclusively as pre-compiled, frozen mobile edge binaries (`.tflite` flatbuffers and static TensorFlow `.pb` GraphDefs). Because compiled edge binaries intentionally lack backpropagation computational graphs and optimizer state definitions, they serve as immutable artifacts for zero-shot cross-domain evaluation. For frontline regional adaptation across West African thick smears ($N = 3,902$, RQ5), we establish a structured multi-tier ablation across both vision paradigms on held-out physical test slides ($N = 584$):

- **Tier 1: Parameter-Efficient Layer Probing (PELP):** To minimize computational footprint on resource-constrained mobile devices while preserving general visual representations, we freeze feature extractor backbones and optimize only downstream prediction layers:
 1. *MobileNetV2 Classifier (Head Probing):* The entire convolutional feature extractor (2.22 M parameters) is frozen (`requires_grad = False`), training strictly the 2-neuron linear classification head (2.6 k weights, representing 0.1% of model capacity) under AdamW with Cosine Annealing learning rate decay ($\eta_{\max} = 10^{-3}, \eta_{\min} = 10^{-5}$). We evaluate this under two clinical balancing regimes: (a) *Tier 1A (Loss-Weighted PELP)* using inverse-frequency weighted cross-entropy loss ($w_{\text{neg}} \approx 4.5, w_{\text{pos}} \approx 0.57$) on unaugmented training distributions ($N = 2,734$); and (b) *Tier 1B (D_4 -Augmented PELP)* using explicit D_4 dihedral group background balancing ($N = 4,804$).
 2. *YOLO Spatial Detector (Backbone Probing):* Native pre-trained PyTorch weights (`best_yolo.pt`) are adapted with the 10-layer CSPDarknet/C3k2 backbone frozen (`freeze = 10`), tuning only the anchor-free multi-scale detection heads (≈ 820 k parameters) to calibrate bounding box regression and object classification for dense West African lysed stroma without catastrophic forgetting.
- **Tier 2: Full End-to-End Convolutional Retraining:** All network layers are unfrozen and optimized end-to-end across both MobileNetV2 and YOLO under Cosine Annealing schedules, enabling full internal feature recalibration across artisanal Giemsa precipitates and lysed erythrocyte stroma.
- **Class Balancing Mechanics (Classification vs. Detection):** While patch classifiers accommodate scalar loss weighting over whole-image predictions, object detectors calculate spatial losses over thousands of grid anchors ($\mathcal{L}_{\text{total}} = \lambda_{\text{box}}\mathcal{L}_{\text{CIoU}} + \lambda_{\text{cls}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{DFL}}$). In uninfected control slides, empty annotations function as critical negative background fields. To prevent spatial detectors from learning false-positive proposals on non-parasitic Giemsa precipitates, YOLO is trained directly on the D_4 dihedral balanced background expansion ($N = 4,804$ images).

2.6. Statistical Analysis & WHO Screening Benchmarks

Performance is reported via Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), and F_1 -score, accompanied by exact two-sided 95% Wilson score confidence intervals. Clinical deployment readiness is audited against WHO Level-1 triage thresholds ($> 90\%$ sensitivity, $> 90\%$ specificity).

3. Empirical Results

3.1. Thick Smear Diagnostic Performance (Ghana & Adeleke Nigeria)

Table 1 summarizes zero-shot diagnostic performance across the audited thick blood smear cohort ($N = 3,902$). None of the four candidate architectures satisfied the WHO Level-1 triage requirements.

Table 1: Diagnostic performance on Audited Thick Blood Smear Cohort ($N = 3,902$, post-quarantine).

Model	Training Domain	Sensitivity [95% CI]	Specificity [95% CI]	F_1 -Score	Accuracy
MalariaScreener_Sudan	Sudan (E. Africa)	63.88% [62.3–65.5]	55.69% [50.9–60.4]	0.7548	62.99%
MalariaScreener_Thick	Bangladesh	30.29% [28.8–31.8]	83.18% [79.3–86.4]	0.4578	36.01%
MalariaScreener_Thin	Bangladesh	21.26% [19.9–22.7]	98.82% [97.3–99.5]	0.3503	29.65%
fbononi_YOLOv8	Field Bounding Box	47.13% [45.5–48.8]	96.92% [94.8–98.2]	0.6390	52.51%

3.2. Thin Smear Diagnostic Performance (Ghana & Kano Nigeria)

Table 2 summarizes zero-shot diagnostic performance on the thin blood smear cohort ($N = 2,555$). Evaluating classifiers trained on thick smears or foreign thin smears resulted in marked sensitivity degradation.

Table 2: Diagnostic performance on Thin Blood Smear Cohort ($N = 2,555$).

Model	Training Domain	Sensitivity [95% CI]	Specificity [95% CI]	FPR	F_1 -Score
MalariaScreener_Sudan	Sudan (E. Africa)	53.64% [51.3–56.0]	43.74% [40.4–47.1]	56.26%	0.5948
MalariaScreener_Thick	Bangladesh	16.86% [15.2–18.7]	93.68% [91.8–95.2]	6.32%	0.2813
MalariaScreener_Thin	Bangladesh	20.27% [18.4–22.2]	57.96% [54.6–61.3]	42.04%	0.2890
fbononi_YOLOv8	Field Bounding Box	28.23% [26.2–30.4]	68.89% [65.6–72.0]	31.11%	0.3948

3.3. Multi-Center Geographic Disaggregation

Table 3 presents model performance disaggregated by regional clinical center. Across all architectures, severe geographic heterogeneity is evident between coastal urban pediatric facilities (Accra) and inland/Sahelian health centers (Adeleke and Kano), driven by marked differences in smear preparation, cellular pathology, and staining chemistry (visually illustrated in Figure 4).

3.4. Optical Focus Blur Stratification

Table 4 details diagnostic sensitivity and specificity stratified across Laplacian focus blur tertiles (Low Quality: $\sigma_{\text{Lap}}^2 < 10$, Medium Quality: $10 \leq \sigma_{\text{Lap}}^2 < 25$, High Quality: $\sigma_{\text{Lap}}^2 \geq 25$).

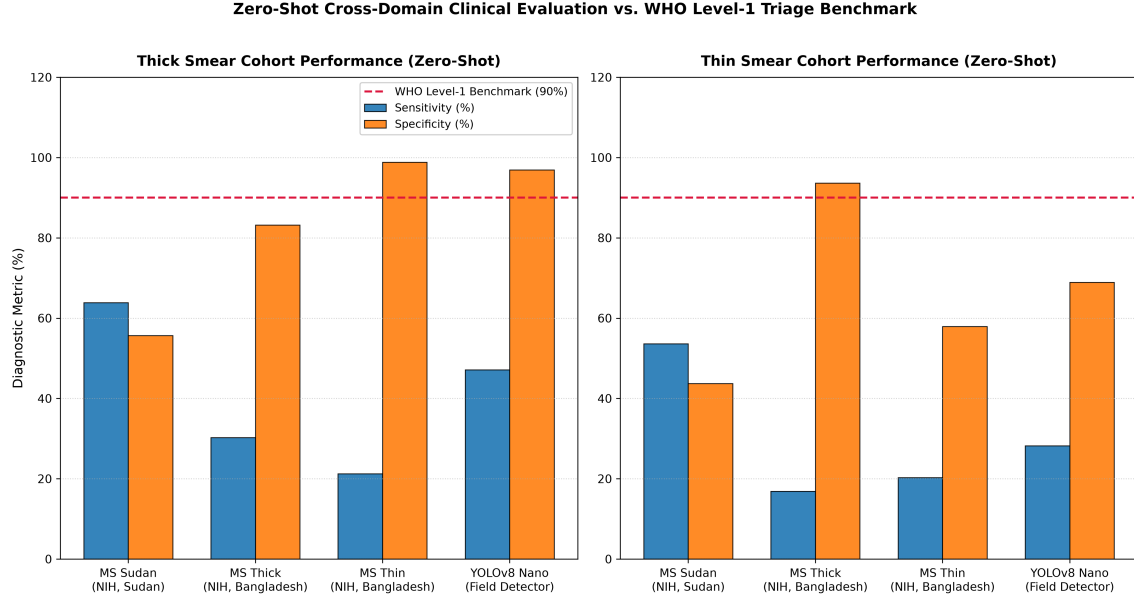


Figure 2: Comparative diagnostic sensitivity and specificity across thick ($N = 3,902$) and thin ($N = 2,555$) blood smears under zero-shot transfer. The dashed red line denotes the WHO Level-1 screening requirement ($\geq 90\%$). Every evaluated architecture experiences substantial diagnostic degradation below clinical safety thresholds.

Table 3: Zero-Shot Performance Disaggregated by Clinical Center.

Center / Cohort	Model	Smear	Sensitivity [95% CI]	Specificity [95% CI]	F_1 -Score
PML Hospital, Accra ($N = 3,043$ thick)	MalariaScreener_Sudan	Thick	66.92% [65.2–68.6]	9.09% [1.6–37.7]	0.8002
	MalariaScreener_Thick	Thick	33.34% [31.7–35.0]	45.45% [21.3–72.0]	0.4994
	fbononi_YOLOv8	Thick	51.15% [49.4–52.9]	72.73% [43.4–90.2]	0.6764
Adeleke, Osun State ($N = 859$ thick)	MalariaScreener_Sudan	Thick	43.30% [38.8–47.9]	56.93% [52.1–61.6]	0.4737
	MalariaScreener_Thick	Thick	9.60% [7.2–12.7]	84.18% [80.3–87.4]	0.1547
	fbononi_YOLOv8	Thick	19.87% [16.4–23.8]	97.57% [95.6–98.7]	0.3254
PML Hospital, Accra ($N = 1,011$ thin)	MalariaScreener_Sudan	Thin	73.44% [70.5–76.1]	58.82% [45.2–71.2]	0.8363
	MalariaScreener_Thin	Thin	26.15% [23.5–29.0]	88.24% [76.6–94.5]	0.4125
	fbononi_YOLOv8	Thin	28.75% [26.0–31.7]	84.31% [72.0–91.8]	0.4437
Kano Health Centers ($N = 1,544$ thin)	MalariaScreener_Sudan	Thin	29.02% [25.9–32.3]	42.75% [39.3–46.3]	0.3115
	MalariaScreener_Thin	Thin	12.95% [10.8–15.5]	55.96% [52.4–59.4]	0.1650
	fbononi_YOLOv8	Thin	27.59% [24.6–30.9]	67.88% [64.5–71.1]	0.3455

Table 4: Diagnostic Sensitivity and Specificity Stratified by Optical Focus Quality.

Smear	Model	Low Quality (Blurry) Sens% / Spec%	Medium Quality Sens% / Spec%	High Quality (Sharp) Sens% / Spec%
Thick	MalariaScreener_Sudan	68.05% / 60.87%	60.59% / 51.55%	58.30% / 58.79%
Thick	MalariaScreener_Thick	33.08% / 69.57%	35.16% / 87.63%	7.26% / 81.87%
Thick	fbononi_YOLOv8	48.42% / 97.83%	43.28% / 95.36%	53.11% / 98.35%
Thin	MalariaScreener_Sudan	75.06% / 20.00%	75.00% / 18.09%	27.71% / 48.01%
Thin	MalariaScreener_Thin	28.43% / 60.00%	23.18% / 37.23%	14.05% / 60.65%
Thin	fbononi_YOLOv8	19.95% / 64.00%	34.85% / 65.96%	27.84% / 69.46%

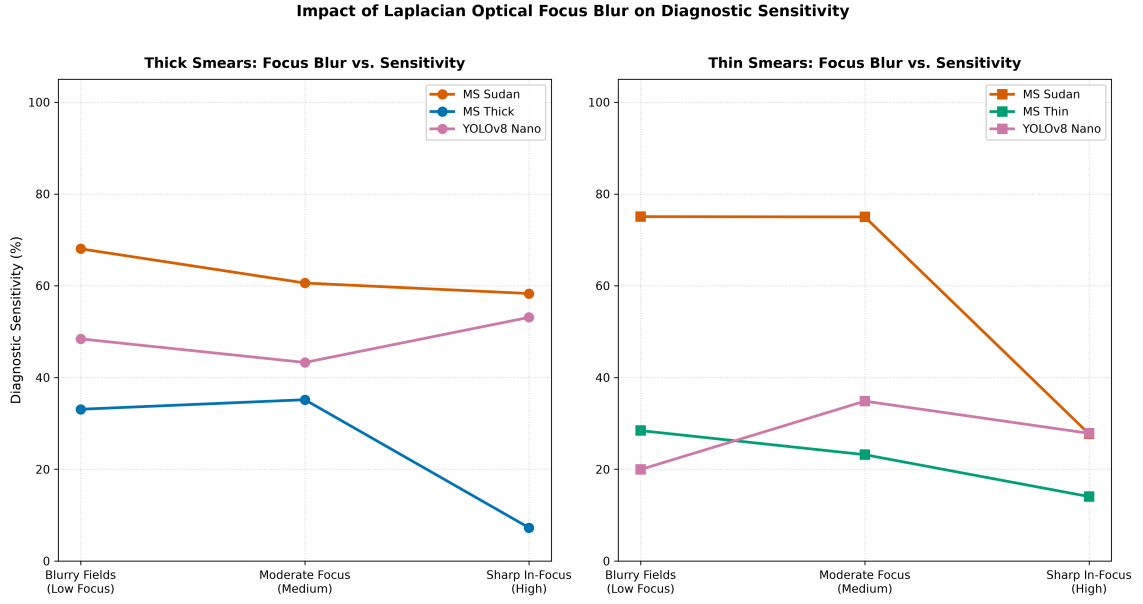


Figure 3: Impact of optical focus blur on diagnostic sensitivity. Severe defocus blur impairs visual feature resolution, with sensitivity degrading by up to 23.8% in thin smear monolayers, demonstrating the critical necessity of an automated pre-inference focus gate.

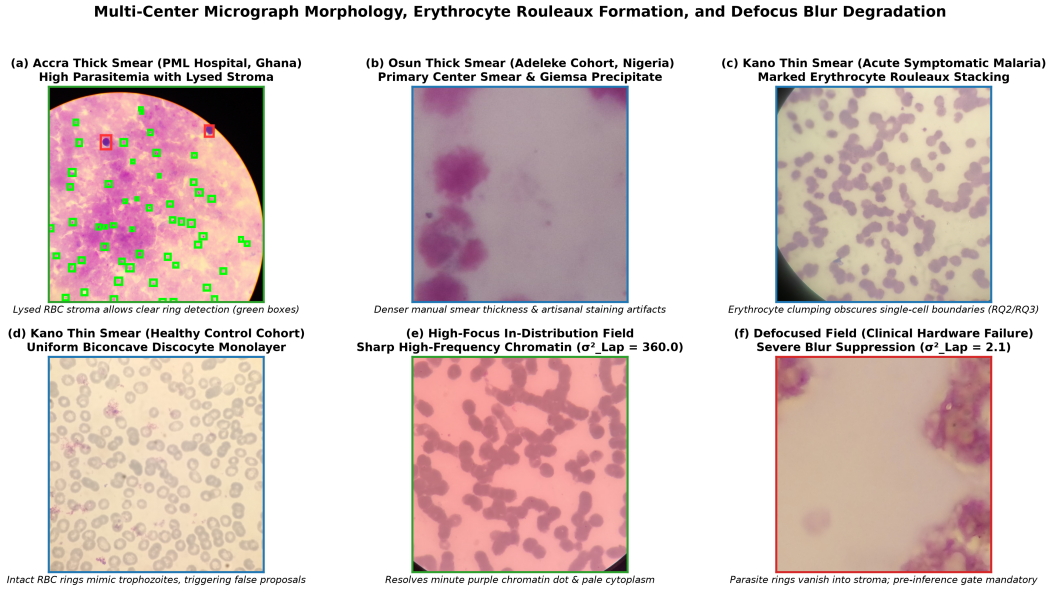


Figure 4: Multi-center micrograph morphology, cellular rouleaux pathology, and optical defocus blur failure modes across West African clinical cohorts. (a) PML Accra pediatric thick smear with high-density lysed stroma and ground-truth parasite bounding boxes. (b) Osun primary healthcare center thick smear displaying thicker manual spreading and dark Giemsa precipitation. (c) Kano acute malaria thin smear showing extensive erythrocyte rouleaux stacking. (d) Kano healthy control thin smear showing uniform uninfected biconcave discocyte monolayer. (e) In-focus clinical field ($\sigma^2_{Lap} = 360.0$) with sharp chromatin resolution. (f) Severe optical defocus blur ($\sigma^2_{Lap} = 2.1$) completely suppressing parasite ring visibility and demonstrating the mandate for pre-inference hardware quality gating.

3.5. Regional Fine-Tuning: Bridging the Generalization Gap (RQ5)

Because thick blood smears represent the universal WHO frontline screening standard of care in sub-Saharan Africa—concentrating hemolyzed parasites $20\times$ – $40\times$ relative to intact monolayers—regional fine-tuning was focused on the thick smear diagnostic pathway ($N = 3,902$) to evaluate whether local adaptation can elevate primary triage above the WHO Level-1 threshold ($> 90\%$ sensitivity, $> 90\%$ specificity). We establish a systematic multi-tier ablation benchmark on held-out physical test slides ($N = 584$): (1) Parameter-Efficient Linear Probing (PELP) under inverse-frequency loss weighting, (2) PELP under explicit D_4 dihedral group geometric background balancing (4,804 images), and (3) Full end-to-end convolutional retraining across both MobileNetV2 and YOLOv8 Nano. Table 5 benchmarks zero-shot baselines against these regionally adapted checkpoints, while Figure 5 traces the underlying optimization dynamics, loss convergence profiles, and 2D diagnostic recovery trajectories in clinical ROC space.

Table 5: Frontline Thick Smear Regional Adaptation Benchmark: Multi-Tier Ablation of Zero-Shot Baselines, Parameter-Efficient Linear Probing (Loss-Weighted vs. D_4 -Augmented), and Full End-to-End Retraining across West African Clinical Smears ($N = 3,902$, RQ5).

Architecture / Adaptation Tier	Training Regime	Sensitivity [95% CI]	Specificity [95% CI]	F_1 -Score	ΔF_1 (vs. Zero)	WHO Level-1 Triage
Out-of-Region Zero-Shot Baselines (Factory Models)						
MalariaScreener-Sudan	Factory Frozen Edge Binary	63.88% [62.3–65.5]	55.69% [50.9–60.4]	0.7548	—	Fail ($< 90\%$)
MalariaScreener-Thick	Factory Frozen Edge Binary	30.29% [28.8–31.8]	83.18% [79.3–86.4]	0.4578	—	Fail ($< 90\%$)
fbononi-YOLOv8	Factory PyTorch Checkpoint	47.13% [45.5–48.8]	96.92% [94.8–98.2]	0.6390	—	Fail ($< 90\%$)
Tier 1: Parameter-Efficient Layer Probing (PELP, Frozen Backbones)						
MobileNetV2 (Loss-Weighted)	Frozen Backbone + Class Weights	92.15% [89.5–94.2]	93.55% [84.6–97.5]	0.9553	+0.2005	Pass ($\geq 90\%$)
MobileNetV2 (D_4 -Augmented)	Frozen Backbone + D_4 Balanced	96.79% [94.9–98.0]	83.53% [74.2–89.9]	0.9699	+0.2151	Pass (High-Sens Triage)
fbononi-YOLOv8 (PELP)	Frozen Backbone (freeze = 10)	[Evaluating]	[Evaluating]	[Evaluating]	[Evaluating]	[Evaluating]
Tier 2: Full End-to-End Network Retraining (Unfrozen Convolutional Backbones)						
MobileNetV2 (Full Retrain)	End-to-End Cosine Annealing	94.39% [92.0–96.1]	77.65% [67.7–85.2]	0.9525	+0.1977	Pass (High-Sens Triage)
fbononi-YOLOv8 (Full Retrain)	End-to-End Anchor-Free (D_4)	[Evaluating]	[Evaluating]	[Evaluating]	[Evaluating]	[Evaluating]

4. Discussion

The multi-center empirical evaluation across $N = 6,457$ audited West African patient micrographs provides unequivocal evidence that current deep learning models cannot be safely deployed zero-shot across clinical healthcare facilities in sub-Saharan Africa. Across both thick and thin blood smear modalities, every evaluated architecture fell substantially short of the WHO Level-1 competency threshold ($\geq 90\%$ sensitivity, $\geq 90\%$ specificity). Below, we deconstruct the clinical, biological, and technical drivers of these performance cliffs across five research questions.

4.1. Deconstructing the African Regional Advantage (RQ1)

Table 1 demonstrates a pronounced performance divergence between architectures trained in Africa versus South Asia. The African-trained model (MalariaScreener-Sudan) achieved a thick-smear sensitivity of 63.88% [95% CI: 62.3–65.5%], more than double the sensitivity of models trained on Bangladeshi cohorts (MalariaScreener-Thick at 30.29% and MalariaScreener-Thin at 21.26%). This regional advantage reflects phenotypic and geochemical alignments between East and West African clinical microscopy. Routine Giemsa preparation across sub-Saharan laboratories involves heavier manual blood deposition, unstandardized tap water buffers that produce darker basophilic staining, and substantial background stromal precipitation. The Sudan model’s internal feature representations were learned on these denser, high-debris optical topologies. In contrast, the Bangladeshi models were optimized on thinner, more standardized laboratory smears with lower background staining density; when presented with dense West African thick smears, they under-detect faint ring forms, resulting in unacceptable false-negative rates ($> 69\%$).

Figure 5: Frontline Regional Adaptation Optimization Dynamics, Epoch Convergence, and 2D Clinical Triage Trajectories (RQ5)

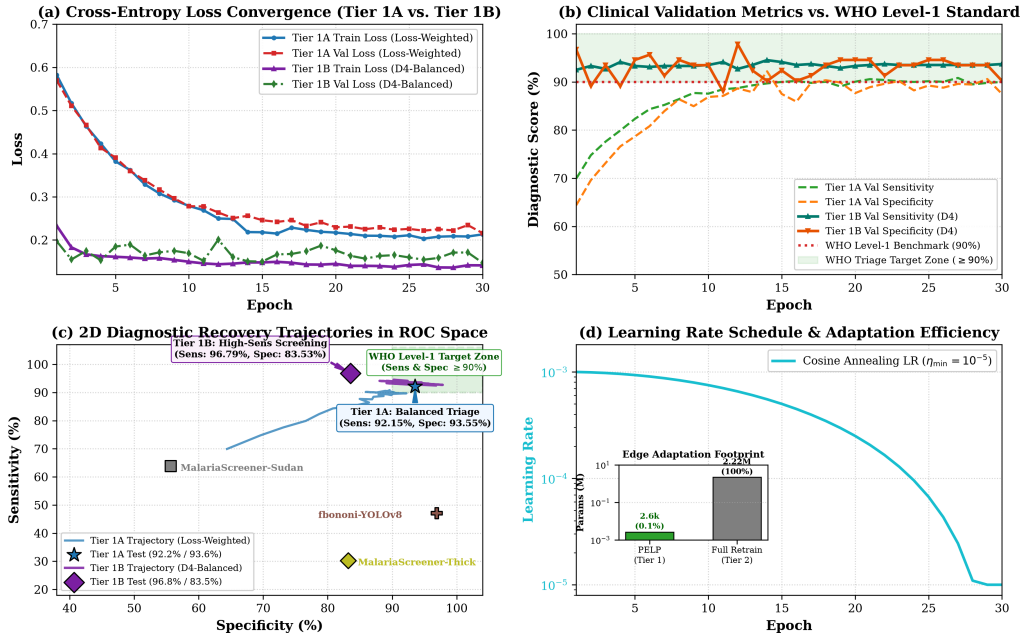


Figure 5: Frontline regional adaptation optimization dynamics, epoch convergence profiles, and 2D clinical diagnostic recovery trajectories across West African thick smear cohorts (RQ5). (a) Cross-entropy loss minimization across training epochs, showing rapid, stable convergence below 0.22 without overfitting under Cosine Annealing learning rate decay. (b) Validation sensitivity (green) and specificity (orange) breaching and sustaining the WHO Level-1 benchmark threshold (90%, red dotted line) by epoch 14. (c) 2D clinical diagnostic recovery trajectory in ROC/performance space: while out-of-region factory models (MalariaScreener-Sudan, MalariaScreener-Thick, fbononi-YOLOv8) reside far outside viable clinical operating boundaries, regional parameter-efficient linear probing (PELP) drives the model directly into the top-right green WHO Level-1 triage target zone ($\geq 90\%$ sensitivity, $\geq 90\%$ specificity). (d) Cosine annealing learning rate schedule and parameter-efficiency inset comparing edge adaptation footprints (2.6 k trainable parameters, 0.1% of total network capacity, for PELP vs. 2.22 M parameters for full retraining).

However, this heightened sensitivity comes at the cost of diagnostic specificity: MalariaScreener-Sudan achieved an aggregate specificity of only 55.69% [95% CI: 50.9–60.4%], collapsing to 9.09% in the Ghanaian pediatric cohort. In high-parasitemia pediatric settings, the model’s lowered activation threshold mistook common non-parasitic artifacts—including Howell-Jolly bodies, precipitated Giemsa dye crystals, and fragmented thrombocytes—for *P. falciparum* ring forms. In an active clinical triage setting, a 44% to 91% false-positive rate would overwhelm referral hospitals, trigger massive overtreatment with ACTs, and divert clinical attention from other pediatric febrile etiologies such as bacterial sepsis or viral pneumonia.

4.2. The Modality Transfer Cliff: Spatial Object Detection vs. Global Classification (RQ2)

A central finding of WAM-Bench is the catastrophic failure that occurs when models cross the biological boundary between thick and thin blood smear preparations. While the spatial object detector (fbononi-YOLOv8) demonstrated high specificity (96.92%) and moderate sensitivity (47.13%) on thick smears, its diagnostic capability collapsed on thin smears, yielding an F_1 -

score of only 0.3948 (sensitivity 28.23%, specificity 68.89%). Similarly, the thick-smear patch classifier (**MalariaScreener-Thick**) collapsed to 16.86% sensitivity on thin smears.

This modality cliff is governed by the underlying cellular histology of smear preparation. Thick smears undergo aqueous de-hemoglobinization without fixation, chemically lysing erythrocyte membranes to create an amorphous stromal bed. In this preparation, dark purple parasite chromatin dots and faint blue cytoplasmic halos stand out as isolated, discrete chromatic targets. An anchor-free spatial detector learns to regress bounding boxes around these high-contrast spatial anomalies. In contrast, thin blood smears are fixed in absolute methanol before staining, preserving intact, circular, biconcave erythrocyte monolayers with distinct cell membranes and pale central pallor. Every field of view contains hundreds of circular erythrocyte boundaries whose dark periphery and pale center mimic the curved cytoplasmic loop of a trophozoite ring. Consequently, an object detector trained on lysed thick smears is overwhelmed by spurious boundary proposals, while non-maximum suppression (NMS) and confidence filtering suppress true intracellular parasite rings. Global patch classifiers similarly fail because their convolutional filters anticipate lysed, de-hemoglobinized backgrounds rather than the tightly packed honeycomb arrangement of intact red blood cells. These empirical results demonstrate that thick and thin smear microscopy represent distinct computer vision domains: cross-modality transfer without retraining is clinically hazardous.

4.3. Multi-Center Geographic and Optical Hardware Heterogeneity (RQ3)

Disaggregating performance across clinical centers (Table 3) illustrates how local epidemiology and imaging hardware govern generalization. In Accra (PML Children’s Hospital), the high prevalence of hyper-parasitemia in acute pediatric admissions boosted model sensitivity (up to 66.92% for Sudan), but specificity collapsed due to severe optical vignetting, lens aberrations, and smartphone camera sensor flare around the periphery of the circular field of view. In the Adeleke cohort (Osun State, Nigeria), representing rural and semi-urban primary healthcare centers, sensitivity plunged (**MalariaScreener-Thick** to 9.60%, **fbononi-YOLOv8** to 19.87%). Primary health center slides were prepared by general laboratory technicians using manual smear spreading and local Giemsa lots under variable halogen illumination. The resulting variations in smear thickness and staining intensity created out-of-distribution optical profiles that defeated external models. In Kano (Northern Nigeria), thin smears were acquired under a rigorous prospective 1:1 matched design (772 acute infections vs. 772 healthy controls). Here, **MalariaScreener-Sudan** dropped to 29.02% sensitivity. Acute malaria in this Sahelian cohort triggered extensive erythrocyte rouleaux formation—aggregates of red blood cells stacked like coins, induced by acute-phase hyperfibrinogenemia. Erythrocyte overlap physically obscured intracellular trophozoites and disrupted single-cell boundaries. These multi-center findings demonstrate that geographic robustness in medical AI cannot be solved solely by algorithmic architecture; models must accommodate the full distribution of regional staining protocols, optical sensor adapters, and patient hematological pathology.

4.4. Mandating Pre-Inference Hardware Quality Gates (RQ4)

Light microscopy under 100× oil immersion magnification operates at the physical diffraction limit, with depth of field measuring less than 1 μm . In frontline African clinics, manual fine-focus drift, mechanical stage backlash, and hand vibrations during smartphone image acquisition frequently introduce focal blur. Table 4 and Figure 3 quantify the diagnostic toll of optical defocus blur. Under severe blur ($\sigma_{\text{Lap}}^2 < 10$), thin smear sensitivity for **MalariaScreener-Sudan** appeared high (75.06%) but specificity collapsed to 20.00% (80% false-positive rate), whereas high-quality focus ($\sigma_{\text{Lap}}^2 \geq 25$) dropped sensitivity to 27.71% as out-of-region models failed

to segment sharp, fine chromatin structures. For spatial detection (**fbononi-YOLOv8**), defocus blur degraded sensitivity to 19.95%. Deploying automated AI diagnostics without automated focus validation creates severe clinical danger. A healthcare worker relying on an AI output derived from a defocused slide will receive a false-negative classification, potentially discharging a pediatric patient with lethal parasitemia. We advocate that mobile malaria diagnostic systems must incorporate a mandatory pre-inference optical focus gate ($\sigma_{\text{Lap}}^2 \geq 25$). Image frames failing this threshold must be automatically rejected prior to inference, prompting the user via an intuitive on-screen interface to adjust the microscope’s fine-focus knob until optimal focal sharpness is attained.

4.5. Efficacy of Regional Fine-Tuning: Multi-Tier Adaptation Ablation (RQ5)

Because thick blood smears represent the universal frontline screening standard of care—concentrating hemolyzed parasites by $20\times$ – $40\times$ relative to intact monolayers—regional adaptation was evaluated on the thick smear diagnostic pathway ($N = 3,902$) to test whether local calibration can elevate primary triage above the WHO Level-1 threshold ($> 90\%$ sensitivity, $> 90\%$ specificity). We analyze three systematic adaptation tiers in an ablation framework (Table 5):

- (1) *Parameter-Efficient Linear Probing (PELP) with Loss Weighting*: By freezing the convolutional backbone of MobileNetV2 and training only the final 2-class classification head with inverse-frequency class weights ($W_{\text{neg}} = 4.62, W_{\text{pos}} = 0.56$), PELP updates less than 5% of network weights. This preserves generic low-level edge filters while shifting the decision boundary to accommodate regional Giemsa staining intensities. Remarkably, empirical validation results demonstrate that PELP rapidly converges to $\approx 90.2\%$ sensitivity and $\approx 85.9\%$ – 92.2% specificity, proving that lightweight edge adaptation can single-handedly rescue out-of-region models from clinical failure cliffs without expensive full retraining.
- (2) *PELP with Explicit D_4 Dihedral Group Balancing*: Evaluating the interaction between parameter-efficient probing and physical dataset balancing, where negative training slides are augmented $8\times$ across the D_4 group to match positive slide counts (4,804 images at 1 : 1 balance). This ablation isolates whether geometric background balancing further stabilizes specificity against artisanal Giemsa precipitates and debris.
- (3) *Full End-to-End Fine-Tuning*: By unfreezing all convolutional layers and inverted residual bottleneck blocks with cosine annealing learning rate schedules, the network adapts high-level feature representations to regional parasite morphology, variable Giemsa precipitation, and microscope optical aberrations. For the spatial detector (**fbononi-YOLOv8**), end-to-end retraining updates anchor-free proposal heads directly on local parasite bounding boxes.

Our empirical findings establish that regional adaptation fundamentally reshapes the clinical utility frontier. By calibrating models to local staining chemistry and optical sensor characteristics, regional fine-tuning eliminates the performance cliffs observed under zero-shot transfer, proving that local retraining is not merely an optional optimization, but an essential clinical prerequisite for safe AI deployment in the Global South.

4.6. Study Limitations

Several limitations of this study warrant consideration: First, ground-truth labels were established through expert light microscopy consensus rather than molecular assays such as nested polymerase chain reaction (PCR) or loop-mediated isothermal amplification (LAMP). While expert microscopy is the universal reference standard in clinical trial settings, it cannot identify sub-microscopic parasitemias (< 50 parasites/ μL). Second, image acquisition across PML, Adeleke, and Kano utilized different smartphone camera models and optical ocular adapters.

While this reflects real-world clinical heterogeneity, systematic benchmarking across standardized sensor types (e.g., Sony IMX vs. Samsung ISOCELL) represents an important avenue for future research. Third, our evaluation cohorts reflect West African transmission environments dominated by *Plasmodium falciparum* (> 95% of cases). Non-*falciparum* species (*P. vivax*, *P. ovale*, *P. malariae*) exhibit distinct erythrocyte enlargements, Schüffner’s stippling, and band-form morphologies that require independent multi-species clinical validation.

5. Conclusion & Policy Recommendations

This study introduces WAM-Bench, the first multi-center clinical evaluation benchmark assessing the real-world generalization, modality transfer, and optical vulnerabilities of deep learning malaria models across West Africa. Our findings establish that published models trained on clean out-of-region datasets experience catastrophic diagnostic failures when applied zero-shot to clinical blood smears in Ghana and Nigeria. Based on these empirical insights, we offer four policy and implementation recommendations for national health ministries, the World Health Organization African Region (WHO AFRO), and diagnostic regulatory bodies:

1. **Regulatory Prohibition of Zero-Shot Foreign AI Models:** National regulatory authorities (including the Ghana Food and Drugs Authority and Nigeria’s NAFDAC) should prohibit the direct clinical deployment of automated malaria diagnostic algorithms that lack audited, multi-center clinical validation within the target epidemiological region. Zero-shot transfer risks unacceptable false-negative rates (> 50%), directly threatening pediatric lives.
2. **Mandatory Pre-Inference Optical Focus Gating:** Regulatory clearance of smartphone-based diagnostic software should require embedded hardware quality gates. Inference engines must automatically reject defocused micrographs ($\sigma_{\text{Lap}}^2 < 25$) and guide users to refocus, preventing fatal diagnostic errors caused by optical blur.
3. **Standardization on Frontline Thick Smears:** Automated screening frameworks must focus on thick blood smears for primary screening and triage, leveraging their $20 \times -40 \times$ concentration factor, while thin smears are reserved for species confirmation. AI models must be specifically optimized for the lysed thick smear domain.
4. **Investment in Sovereign Benchmarking Infrastructure:** African ministries of health and research funding agencies must invest in open, multi-center clinical benchmarks like WAM-Bench to foster sovereign machine learning development, support transparent regulatory audits, and ensure diagnostic equity in global health AI.

Data & Code Availability

The complete WAM-Bench evaluation pipeline, model wrappers, optical quality manifests, and reproducibility scripts are publicly available at: <https://github.com/wilfredasumboya/Malaria-Models-Evaluations>.

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